MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6DC6

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Pentode, miniature, rf

Outline --- 5-2 (EIA)

Base --- E7-1

Envelope --- T5-1/2

Cathode --- Coated unipotential

Base connections:

Pin no.	---	1	2	3	4	5	6	7
Element	---	g1	k	h	h	a	g2	g3, int sd

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ecc2	Ec2	Ec3	Ec1	Ehk
Unit:	V	Vdc	Vdc	Vdc	Vdc	Vdc	v
Maximum:	6.9	330	330	165	0	0	200
Minimum:	5.7	---	---	---	---	---	---
TEST CONDITIONS:	6.3	200	---	150	Note 1	0	Note 2

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Rk	Rg	Pp	Pg2	TE	Alt
Unit:	Ohms	Meg	W	W	°C	ft
Maximum:	---	0.25	2.2	0.55	140	Note 4
Minimum:	---	Note 3	---	---	---	---
TEST CONDITIONS:	180	---	---	---	---	---

GENERAL:

Qualification - Required

(B) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1266	Total grid current	5	Rg1 = 0.25 Meg	0.65	II	Ic1	---	-1.0	µAdc
1256	Electrode current (anode)	-		0.65	II	Ib	6.0	12.0	mAdc
1306	Transconductance (1)	-		0.65	II	Sm	4,000	7,000	µmhos
1306	Transconductance (2)	-	Ec1 = -10 Vdc	0.65	II	Sm	10	300	µmhos
1201	Short and discontinuity detection	-		0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
(B) 1246	Audio frequency noise	-	Ebb = 200 Vdc; Ec2 = 150 Vdc; Ck = 50 µF; Rb = 10,000 ohms; Rg1 = 0.1 Meg; Esig = 150 mVac	2.5	I	EB	---	17	vu
1256	Electrode current (screen)	-		---	---	---	---	4.3	mAdc
1211	Insulation of electrodes	-		---	---	---	---	---	---
1301	Heater current	-		---	---	If	275	325	mA
1336	Heater-cathode leakage	-		---	---	Ihk	---	20	µAdc
1306	Transconductance (3)	-	Ef = 5.7 V	---	---	Sm	3,800	---	µmhos
1331	Direct-interelectrode capacitance	-	No external shield	6.5	Code E	Cin Cout Cg1p	4.7 1.5 ---	7.3 2.5 0.024	pF pF pF
1121	Base strain	-		---	---	---	---	---	---
2126	Glass strain	-		2.5	I	---	---	---	---
1105	Permanence of marking	-		---	---	---	---	---	---
	Quality conformance inspection, part 3								
1501	Intermittent life	-	Group A; Ehk = +180 Vdc; Eb = 200 Vdc; Ec2 = 150 Vdc; Rk = 150 ohms; Rg1 = 1.0 Meg	---	---	---	---	---	---
---	Intermittent life-test end point (500 hours):								
1306	Transconductance (1)	-		---	---	Sm	3,200	---	µmhos

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NOTES:

1. Tie grid No. 3 to positive terminal of cathode resistor.
2. The absolute value of the dc heater-cathode voltage shall not exceed 100 volts.
3. This value is for operation under fixed-bias conditions. With cathode bias Rg1 may be 1.0 megohm maximum.
4. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
5. This test to be performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 11, 17
DSA - ES

User activities:
Army - WC
Navy - AS, OS, MC, CG, SH

Preparing activity: Army - EL

Agent: DSA - ES

(Project 5960-2621)

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Department of the Army
Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding General
U.S.Army Electronics Command
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Fort Monmouth, New Jersey 07703

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-E-1/1568B ELECTRON TUBE, RECEIVING TYPE 6DC6

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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